Sampling & Estimation

ANSWER KEY

Section 1: Unbiased Estimators of Population Mean & Variance

1 What is the unbiased estimator of the population variance σ^2 calculated from a random sample of size n?

a. $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ ✓

b. $s^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum(x_i - \bar{x})^2$

d. $s^2 = \sum(x_i - \bar{x})^2$

Dividing by $n-1$ instead of n removes bias, making $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ the unbiased estimator of σ^2.

2 A random sample of 10 observations has $\sum x = 80$ and $\sum x^2 = 724$. Calculate the unbiased estimate of the population variance.

a. 9.33 ✓

b. 8.40

c. 10.50

d. 7.20

$s^2 = \frac{1}{n-1}\left[\sum x^2 - \frac{(\sum x)^2}{n}\right] = \frac{1}{9}\left[724 - \frac{6400}{10}\right] = \frac{1}{9}[724 - 640] = \frac{84}{9} \approx 9.33.$

3 If the biased sample variance is $v = 18$ from a sample of size $n = 25$, find the unbiased estimate s^2.

a. 18.75 ✓

b. 17.28

c. 18.00

d. 19.50

$s^2 = \frac{n}{n-1}v = \frac{25}{24}(18) = 18.75.$

Section 2: The Central Limit Theorem (CLT)

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4 State the Central Limit Theorem for the distribution of the sample mean $\bar{X}$.

a. **For large n, $\bar{X}$ is approximately normal $N(\mu, \sigma^2/n)$ regardless of the population distribution** ✓

b. For any n, $\bar{X}$ is strictly normal

c. For large n, the population becomes normal

d. $\bar{X}$ has mean $n\mu$ and variance $n\sigma^2$

The CLT states that the sample mean $\bar{X}$ of a random sample from ANY distribution with mean μ and variance σ^2 is approximately normally distributed $N(\mu, \sigma^2/n)$ as $n \to \infty$ (typically $n \geq 30$).

5 A population has mean 100 and standard deviation 15. A sample of size $n = 36$ is taken. What is the standard error of the sample mean?

a. 2.5 ✓

b. 15

c. 4.17

d. 0.417

Standard error $\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{36}} = \frac{15}{6} = 2.5$.

6 Using the distribution from the previous question ($\mu = 100, \text{SE} = 2.5$), find $P(\bar{X} > 105)$.

a. 0.0228 ✓

b. 0.0500

c. 0.0456

d. 0.0100

$z = \frac{105 - 100}{2.5} = 2.0$. $P(\bar{X} > 105) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228$.

Section 3: Confidence Intervals for the Population Mean

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7 A sample of size $n = 64$ from a normal population with known $\sigma = 8$ has sample mean $\bar{x} = 52.0$. Find the 95% confidence interval for μ.

a. $[50.04, 53.96]$ ✓

b. $[49.50, 54.50]$

c. $[50.36, 53.64]$

d. $[51.00, 53.00]$

Margin of error $= z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{8}{8} = 1.96$. Interval $= 52.0 \pm 1.96 = [50.04, 53.96]$.

8 What is the critical value z used to construct a 99% confidence interval for the population mean?

a. 2.576 ✓

b. 1.960

c. 1.645

d. 2.326

For a 99% two-tailed confidence interval, $1 - \alpha = 0.99 \Rightarrow \alpha/2 = 0.005$. $\Phi(2.576) = 0.995$.

9 How large must a sample be so that the width of a 95% confidence interval for μ is at most 2 units, given $\sigma = 10$?

a. **At least 385** ✓

b. At least 100

c. At least 196

d. At least 250

Width $= 2z\frac{\sigma}{\sqrt{n}} \le 2 \Rightarrow z\frac{\sigma}{\sqrt{n}} \le 1 \Rightarrow 1.96 \times \frac{10}{\sqrt{n}} \le 1 \Rightarrow \sqrt{n} \ge 19.6 \Rightarrow n \ge 19.6^2 = 384.16$. Least integer is 385.

Section 4: Confidence Intervals for Large Unknown-Variance Samples

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10 When σ^2 is unknown but the sample size $n = 100$ is large, what is substituted for σ^2 in calculating the confidence interval?

a. **The unbiased sample variance estimate** s^2 ✓

b. The sample range divided by 4

c. The median absolute deviation

d. A default value of 1.0

By the Central Limit Theorem and Slutsky's theorem, when n is large, the unknown population variance σ^2 is replaced by the unbiased sample variance estimate s^2.

11 A sample of 100 lightbulbs has mean life $\bar{x} = 1200$ hours and unbiased sample variance $s^2 = 2500$. Find the 90% confidence interval for the true mean life.

a. $[1191.8, 1208.2]$ ✓

b. $[1180.0, 1220.0]$

c. $[1190.2, 1209.8]$

d. $[1195.0, 1205.0]$

$s = \sqrt{2500} = 50$. Standard error $= 50/\sqrt{100} = 5$. For 90% confidence, $z = 1.645$. Margin $= 1.645(5) = 8.225$. Interval is $[1191.8, 1208.2]$.

12 What is the correct probabilistic interpretation of a 95% confidence interval $[L, U]$ for μ?

a. **In repeated sampling,** 95% **of such constructed intervals will contain the true parameter** μ ✓

b. There is a 95% probability that μ lies in this specific interval

c. 95% of population values fall between L and U

d. The sample mean falls in $[L, U]$ 95% of the time

The parameter μ is a fixed unknown constant; the interval endpoints are random variables. 95% of intervals generated by this procedure over repeated samples will capture μ.

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